higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

2 December 2019 (X-Paper)
09:00–12:00

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This question paper consists of 6 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 A closed-loop controller requires an input from both the process variable and the …

A digital variable.
B analogue variable.
C set point.
D error signal.

1.1.2 When a process is balanced, process variable …

A is set point.
B is greater than set point.
C is less than set point
D is 50% of set point

1.1.3 Which ONE of the following pH values indicates a basic solution?

A 0
B 4
C 7
D 11

1.1.4 In oxidation reduction reactions, a substance giving up the electrons is called a/an …

A reducing agent.
B oxidant.
C oxidising agent.
D None of the above

1.1.5 Which ONE of the following function does a primary device perform in the indirect measurement of fluid flow?

A Measures the pressure difference
B Create the pressure difference
C Measure the amount of flow
D Assures a smooth flow

(5 × 1) (5)

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1.2 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–I) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	A measure of water vapour present in the air	A	density
1.2.2	The temperature at which the water vapour present in the air saturates the air and begins to condense	B	saturation temperature
1.2.3	The relationship between the viscosity of gases and liquids to the steady state velocity of a known weight sphere, falling through a fluid of known density at low Reynolds number	C	non-newtonian fluids
1.2.4	When fluids are deformed by strain, the ratio between shear rate and shear stress will not be constant value	D	humidity
1.2.5	An apparatus that measures the amount of water vapour in the air	E	Newtonian fluids
		F	hygrometer
		G	dew point
		H	hydrometer
		I	Stoke's law

(5 × 1) (5)

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1.3 Complete the following sentences by using the words in the list below. Choose the answer and write only the answer next to the question number (1.3.1–1.3.3(b)) in the ANSWER BOOK

voltage; volumetric; square root; velocity; differential pressure

1.3.1 Turbine and magnetic flow meters are … flow measuring devices. (1)

1.3.2 The output (a) …of the magnetic flow meter is proportional to the (b) … of the fluid moving through the meter. (2)

1.3.3 All Bernoulli type primary flow elements set up a (a)…. which varies as a (b) …of the flow rate. (2)

[15]

TOTAL SECTION A 15

SECTION B

QUESTION 2: FLOW MEASUREMENT

2.1 Prove with the aid of the energies associated with the liquid flowing in a pipe and the application of Bernoulli's theorem for liquids, that $Q = A_2 . E\sqrt{2gh}$ for orifice plate. (15)

2.2 Name and explain the correcting factors that must be included in practice in the basic equation in QUESTION 2.1 in order that the theoretical values matches the practical values. (6)

2.3 A V–notch weir is installed in a channel to maintain a flow rate of 0,7 m^3/s.

If the notch is manufactured at 50°, calculate the head developed in the weir. Take g = 10 m/s^2. (4)
[25]

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY

3.1 Make a neat labelled sketch of a balanced flow vessel for measuring specific gravity and density. (5)

3.2 Explain the working principle of the hydrometer in QUESTION 3.1. (5)

3.3 Explain with the aid of a sketch, the working principle of a capillary-tube viscometer. (15)
[25]

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QUESTION 4: pH MEASUREMENT

4.1 What is the necessity, should the pH sensitive electrode be operating in or near an electrical field? (2)

4.2 Why is screening and earthing important in pH measurement? (3)

4.3 Explain in details, what you understand by a pH meter registering seven on its scale. (2)

4.4 Define the term *isopotential point.* (2)

4.5 Describe the process involved in the cleaning of pH electrodes. (6)
[15]

QUESTION 5: AUTOMATIC CONTROL

5.1 Complete the following sentences by using the words provided in the list below. Write only the word(s) next to the question number (5.1.1(a)–5.1.5(d)) in the ANSWER BOOK.

last position; equilibrium; control element ; process variable; percentage ; desired value; corrective signal; slightly beyond; inactive neutral; three or less; pre-set

5.1.1 Position control is a mode of control in which the final (a) … , is quickly moved to one of a few positions, usually (b) … when the measured variable reaches a predetermined magnitude. (2)

5.1.2 Two position with an (a) … is one in which the final control element remains in its (b) … until the measured variable has moved (c) … its control point. (3)

5.1.3 Inherent control is that without a controller, the system reaches … after a disturbance. (1)

5.1.4 Gain is the … proportional band. (1)

5.1.5 Closed loop control is a combination of control units in which the (a) … is measured and compared with the (b) ... If the measured value differs from set point, (c) …, is sent back to the final control element to bring back the controlled variable to the (d) … value. (4)

52 Make a neat labelled sketch of a two-term (proportional + integral) electronic controller using an operational amplifier. (5)

5.3 Sketch a block diagram of a cascade control loop. (4)

[20]

TOTALSECTION B: 85
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

(1) $$Q = \frac{8}{15}\tan\left(\frac{\theta}{2}\right)\sqrt{2gH^5}$$

(2) $$Q = \frac{2}{3}B\sqrt{2gH^5}$$

(3) $$pH = -\log[H^+]$$

(4) $$.Q = 0{,}01252.N.D^2.\sqrt{\frac{h}{\rho}}$$

(5) $$m = \left(\frac{d}{D}\right)^2$$

(6) $$R_D = \frac{3{,}54.Q\rho}{\mu d}$$

(7) $$E = \frac{1}{\sqrt{1-m^2}}$$

(8) $$\frac{h}{P_{abs}} = \frac{\Delta P}{P_g + P_{atm}} \times 27{,}2$$

(9) $$Q = 0{,}01252 Z_R Z_D \epsilon C E d^2 \sqrt{\frac{h}{\rho}}$$

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